TRAUMATIC INJURIES MODULE

What to remember

Online Module Quick Reference

Each lesson of the online module is distilled to its four or five essential ideas. This document does not replace the manual, the interactive capsules or in-class practice. It gives you a reference point before the in-person session and a quick refresher after the course.

Traumatic injuries are soft tissue injuries that require specific care because they involve the head, face, neck, spine, chest or abdomen: regions where the visible injury often hides the one that matters. The online module lays out the anatomy and the particularities of each region; assessment and management of head, spine, chest and abdominal injuries are worked on in class.

MODULE CONTENTS: TEN LESSONS

00 Introduction to the chapter

Injuries where the visible wound often hides the one that matters

Why this chapter: a scalp bleed is impressive, a fractured rib hurts, a broken nose is obvious. None of the three kills. What kills is pressure inside the skull, a collapsing lung, a bleeding spleen. Here you learn to look behind the injury.

1. **Three blocks: face and head, nervous system and spine, trunk.** The first five lessons cover the structures of the face. The next three lay out the anatomy of the nervous system, spinal motion restriction devices and the particularities of children. The last prepares for the chest and abdominal injuries covered in class.
2. **One reflex for this whole chapter: the airway.** Blood in the mouth, knocked-out tooth, neck swelling, unconscious patient: the first threat from a head or facial injury is almost always obstruction. Recovery position as soon as blood or vomit threatens.
3. **A second reflex: the cervical spine.** Any significant head or facial trauma leads to a presumption of cervical injury until proven otherwise. The spine algorithm, worked on in class, tells you when you can lift that presumption.
4. **Which type of shock?** Neurogenic for a high cervical injury, hypovolemic for a bleeding abdomen or chest, respiratory for a compressed lung. The question follows you from one lesson to the next.
5. **Anatomy online, assessment in class.** Head injuries, spinal injuries, SMR, chest and abdominal injuries: these management skills are taught and practised in person, with the decision algorithm and the scenarios.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 8, Traumatic Injuries · Online encyclopedia	**Pages 105 to 127, traumatic injuries section**	**Head injuries, spine and SMR algorithm, chest, abdomen, three-patient rockslide scenario**

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01 Facial and neck injuries

A methodical approach to fragile and vital structures

> **The face: eight bones, four sense organs, the entrance to the airway.** The neck: the trachea, the esophagus, the carotids, the jugulars, and the cervical spine that supports it all. An injury here can compromise breathing, the skull and the spine at the same time.

1. **Three questions before treating the visible injury.** Is the airway threatened (edema, blood, clots, tooth or denture)? Is the skull involved? The cervical spine? A serious facial trauma answers yes to all three until proven otherwise.
2. **The exam: look for asymmetry, then palpate.** Rim of the orbits, maxillary bones, mandible: deformity, crepitus, air bubbles under the skin (sinus fracture). Mouth: bleeding, loose tooth, full opening, normal bite ("do the teeth fall into place?"). A jaw that does not close properly is a fractured jaw.
3. **Management of the face.** Moderate compression (facial bones are fragile), ice on contusions but never on the eyes, removal of objects impaled in the cheek with pressure inside and out, recovery position, no airway adjunct in cases of significant trauma to the mouth or nose.
4. **The neck: direct compression, never a circumferential bandage.** Puncture wound: sterile gauze and gloved fingers on the site, held by a diagonal bandage under the opposite arm. Ruptured artery: compression above and below, maintained throughout the evacuation. A bandage around the neck strangles.
5. **Penetrating neck trauma: bleeding takes priority over the spine.** Stop the hemorrhage first; do not use a cervical collar, rigid or improvised. Immediate evacuation.

LEARNER'S MANUAL	FIELD MANUAL	EN LIGNE
Section 8.1, Facial Injuries · Online encyclopedia	**Pages 105 and 106, facial trauma**	**ThingLink scene: physical exam of the face and neck**

02 Eye injuries

Recognize, protect, and know when to do nothing

The eye is protected by the orbit and the eyelid, and that is all. A foreign body is the most frequent eye injury in the outdoors; an impaled object and blunt trauma are the ones that threaten vision.

1. **Foreign body: look before acting.** Remove contact lenses. Evert the upper eyelid over a cotton swab, use grazing light from the outer side to reveal the particle or the scratch. Irrigate from the nose side toward the ear (drinking water or ophthalmic solution), remove with a cotton swab moistened with ophthalmic ointment. Never rub.
2. **Impaled or embedded object: remove nothing.** Stabilize with a gauze ring, cover the injured eye, and cover the uninjured eye too, leaving a small opening in the centre to limit movement of both eyes. Immediate evacuation. Same rule for an enucleated eye, covered with a moist dressing.
3. **Blunt trauma: cover both eyes, evacuate sitting up.** Hyphema (blood in front of the iris), orbital fracture, retinal detachment: vision decreases or distorts. No ASA, which promotes bleeding. Walking limited to what the evacuation requires.
4. **Snow blindness: a sunburn of the cornea.** Pain, gritty sensation, tearing, 6 to 12 hours after exposure. Rest in darkness, cool compresses, healing in 24 to 48 hours. Prevention: glasses with side protection, even in overcast weather.
5. **Abrasion and infection: relieve and avoid spreading.** A corneal abrasion heals in 24 hours; an eye patch helps with comfort. Infection (redness, pus in the morning): irrigation, antibiotic ointment or drops in both eyes, no shared towel. Only products intended for the eyes go in the eyes.

LEARNER'S MANUAL
Section 8.2, Eye Injuries · Online encyclopedia

FIELD MANUAL
Pages 107 to 109, the eyes

EN LIGNE
Video: everting the upper eyelid

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TRAUMATIC INJURIES

03 Nose injuries

Bleeding, fracture and sinusitis

Anterior nosebleed is almost always benign. What changes everything is the mechanism: after a significant blow to the head, blood or clear fluid running from the nose is a sign of head injury, and you do not stop it.

1 Sit up Head tilted slightly forward, mouth open	**2 Pinch 15 min** Both nostrils, without letting go to check	**3 Cold** Cold compress on the bridge of the nose	**4 Pack, then evacuate** Gauze in the nostril if it persists; evacuate if it recurs or blood loss is significant

Figure 1. Epistaxis without head injury or cervical injury.

1. **With or without head injury: the first question.** Significant blow to the head: watch for signs of head injury and cervical injury, do not try to stop the flow (it increases intracranial pressure), watch for blood in the airway.
2. **Simple epistaxis: the sequence above.** Continuous pinching for at least 15 minutes, by the patient themselves if possible. Have them spit rather than swallow: swallowed blood causes nausea and masks the volume lost.
3. **Nasal fracture: ice, no reduction.** Deformity, tenderness on palpation, bleeding. Ice for swelling, bleeding control, examination of the neighbouring structures (orbits, sinuses). Reduction is a medical procedure; evacuation is urgent only if the bleeding cannot be controlled or the airway is threatened.
4. **Sinusitis: decongest, no antibiotics.** Inflammation of the cavities around the nose, most often viral or allergic: facial pressure, headache or earache, discharge. Saline spray, hydration, analgesic. The mucus must drain.

LEARNER'S MANUAL
Section 8.3, Nose Injuries · Online encyclopedia

FIELD MANUAL
Page 112, nasal conditions

04 Ear injuries

Lacerations, foreign bodies, eardrum and infections

The ear is for hearing and for staying upright. Outer (pinna, canal, eardrum), middle (ossicles, connected to the nose by the Eustachian tube), inner (balance). An inner ear injury shows up as vertigo as much as hearing loss.

1. **Laceration and avulsion of the pinna: like a wound, with padding.** Direct compression, careful cleaning, flap repositioned before dressing. Folded gauze pads between the ear and the scalp hold the dressing without crushing the pinna.
2. **Foreign body: irrigate with lukewarm water, ear facing the ground.** Insect, sand, debris: irrigation flushes them out. Never insert a pointed object into the canal.
3. **Perforated eardrum: cover, insert nothing.** Blast effect (lightning, avalanche), penetrating object, or infection. Hearing loss, tinnitus, loss of balance. Dry sterile gauze over the ear, impaled object stabilized, no packing, evacuation.
4. **Blood or clear fluid draining from the ear: head injury.** Do not stop it, do not plug it. It is a sign of a basal skull fracture; it changes the evacuation priority.
5. **Swimmer's ear: keep the ear dry.** Redness, warmth, swelling and pain of the canal after prolonged immersion. Antibiotic ear drops, gentle cleaning, ears kept dry. Viral earaches are treated with the analgesics and antihistamines used for colds.

LEARNER'S MANUAL
Section 8.4, Ear Injuries · Online encyclopedia

FIELD MANUAL
Page 113, ear laceration

TRAUMATIC INJURIES

05 Dental injuries

Avulsion, fracture and lost filling

A knocked-out tooth is reimplanted, and quickly. The chances of success drop after 30 minutes. But before the tooth, a mouth injury leads to a presumption of head injury and cervical injury, and threatens the airway.

1. **First the mouth, not the tooth.** Airway, position that keeps blood from running toward the throat, inspection: loose or knocked-out teeth, displaced denture, foreign bodies. An exposed nerve is very painful: cover it with gauze to protect it from the cold.
2. **Avulsion: rinse, align, replace.** Handle the tooth by the crown, rinse without scrubbing, compare with the opposite tooth to orient it, replace it in its socket in alignment. Unpleasant, rarely very painful. Evacuation to a dentist as quickly as possible.
3. **If you cannot reimplant: keep it moist and cool.** Conscious patient: between the cheek and the teeth. Unconscious patient or risk of aspiration: moist sterile dressing, saline solution or milk. Never dry.
4. **Fracture or lost filling: protect and relieve.** Dry the tooth, avoid contact with the tongue and saliva, topical analgesic (benzocaine 20%) on the gum, dental wax or temporary filling material on the broken edges or the cavity. Never temporary filling material on an abscess or an infected socket: the pressure would increase.
5. **Plan a dental kit for long trips.** Wax, temporary filling material, topical analgesic, mirror, gauze. Untreated dental pain can end an expedition as surely as a fracture.

LEARNER'S MANUAL

Section 8.5, Dental Injuries · Online encyclopedia

FIELD MANUAL

Page 111, broken tooth and filling

TRAUMATIC INJURIES

06 Anatomy and functions of the nervous system

What you need to understand to recognize a head or spinal injury

The central nervous system commands, the peripheral transmits. The brain and spinal cord are enclosed in the skull and the spine, bathed in cerebrospinal fluid, in a space that cannot expand. That is what explains intracranial pressure and the severity of spinal cord injuries.

1. **The skull is a closed box.** Brain, blood and cerebrospinal fluid share a fixed volume. Anything added (bleeding, edema) compresses the brain. Signs to monitor for 24 hours: worsening headache, projectile vomiting, altered level of consciousness, unequal pupils, changing respiration and pulse.
2. **Skull fracture and brain injury do not always go together.** One can exist without the other. Blood or clear fluid from the nose or ears, bruising behind the ears or around the eyes (late signs) suggest a basal skull fracture.
3. **The spine protects the cord, region by region.** Seven cervical vertebrae (the most mobile, they carry the head), twelve thoracic (the most stable, held by the ribs), five lumbar (the most loaded, the most often injured), then the sacrum and the coccyx. The cord ends at L2; the nerves continue lower.
4. **Central and peripheral, somatic and autonomic.** The CNS (brain, cord) coordinates; the PNS (sensory and motor nerves, reflexes) connects to the rest of the body. The autonomic system regulates without conscious control: the sympathetic speeds up (pulse, respiration, vasoconstriction), the parasympathetic slows down. A cervical injury can cut off this control.
5. **Two complications of a spinal injury.** Breathing difficulty if control of the diaphragm is affected: the first aider must manage the airway and ventilate. Neurogenic shock for injuries up to T6: vasodilation with hypotension and slow pulse, the opposite of hypovolemic shock.

LEARNER'S MANUAL
Sections 8.6 and 8.7 (anatomy) · Online encyclopedia

FIELD MANUAL
Pages 114 to 120, head injury, mild TBI and spinal trauma

IN CLASS
Assessment of head injuries, red flags and evacuation, spinal injuries and SMR algorithm

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07 Spinal immobilization devices

Restricting movement during transport, with what you have

Spinal motion restriction (SMR) protects the cord by limiting movement during transport, without prolonged mechanical immobilization. Backboard and rigid collar are options, not the rule.

Device	Use	Limit
Vacuum mattress	Long transport, moulds to the body	Bulk, rarely in a kit
Basket stretcher	Difficult terrain, helicopter hoist	Padding and strapping need attention
Scoop stretcher	Pickup without rolling the patient	Transfer, not long transport
KED	Extrication of a seated patient	Extrication only (SIRIUSMEDx choice)
Backboard	Transfer platform	Never a long-term tool
Cervical collar	Reminder not to move	Unnecessary if other support; never in penetrating neck trauma

Figure 2. The devices, their use and their limit.

1. **The vocabulary.** Spinal precautions: all the measures that limit movement in a patient at risk. SMR: the measures used during transport, whatever the method. Mechanical immobilization: with a device. Extrication: the phase before the definitive litter.
2. **Choose according to three factors.** The means available, the evacuation time, the patient's condition. Alert and reliable: they can move themselves while limiting neck movement. Unconscious: you protect the spine and manage the airway.
3. **The principles that do not change.** Neutral position without forcing, stop if there is resistance, padded surfaces, straps that restrict neither breathing nor circulation, head secured last, extremities accessible for CSM, continuous monitoring.
4. **Transfer by BEAM rather than by log roll.** Lift and slide as a unit, cervical stabilization with the trapezius squeeze. Side-lying position is acceptable if the patient cannot tolerate lying on their back, is vomiting, or if constant monitoring is impossible.
5. **If you cannot ensure SMR: wait.** For a high-risk patient, barring a life-threatening emergency, wait for search and rescue or air medical transport rather than a transport that makes things worse.

LEARNER'S MANUAL
Section 8.7, Transport and Stabilization Devices · Online encyclopedia

FIELD MANUAL
Pages 121 to 123, transport, vacuum mattress, backboard

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Videos and photos: not all devices can be seen in class; optional vacuum mattress video

08 Pediatric considerations

Assessing and managing a spinal injury in a child

> **A child is not a small adult.** Their head is proportionally larger, their ligaments looser, their vertebrae still cartilaginous. They compensate for a long time, then decompensate quickly. And the adult decision rules have not been validated in children.

1. **The large head flexes the neck when lying flat.** In a young child, the prominent occiput brings the neck into flexion when lying flat on the back. To achieve neutral position, raise the torso and shoulders, not the head.
2. **An injured cord without a visible fracture.** Ligament laxity allows the cord to stretch without bony injury. A neurological deficit, even transient (weakness, tingling after the fall), counts as an injury, even if the child seems fine afterward.
3. **The decision rules do not apply as is.** The algorithm criteria (NEXUS, Canadian rule) were validated in adults, to decide on an X-ray. In children, the reliability of the exam (pain, orientation, cooperation) is often insufficient: when in doubt, apply spinal precautions.
4. **The mechanism carries more weight than in adults.** Fall from a height greater than their own height, collision, ejection, diving, animal bite to the face or neck: presume a cervical injury. A child compensates for shock until losing a large part of their blood volume, then collapses.
5. **Adapt motion restriction to their size.** Adult devices do not hold them: fill the gaps with clothing or foam, use the parent to calm and keep the head still, monitor breathing closely (the diaphragm does almost all the work in small children).

LEARNER'S MANUAL	FIELD MANUAL
N/A	N/A

09 Anatomy of the chest and abdomen

What the two cavities contain, and what bleeds or collapses

Two cavities separated by the diaphragm. The chest contains what breathes and what pumps, protected by the ribs. The abdomen contains what digests, filters and stores blood, protected by almost nothing. The exam is done by four quadrants.

Quadrant	Main organs	Injury to suspect
Upper right	Liver, gallbladder, right kidney	Bleeding liver: massive internal hemorrhage
Upper left	Spleen, stomach, left kidney	Ruptured spleen: hypovolemic shock, sometimes delayed
Lower right	Appendix, intestine, colon	Perforation of a hollow organ: peritonitis
Lower left	Intestine, sigmoid colon	Perforation, hemorrhage

Figure 3. Four quadrants, and the injury to suspect in each.

1. **The chest: negative pressure and pleura.** Twelve pairs of ribs, the sternum, the diaphragm. The lungs are wrapped in an airtight pleura; it is the negative pressure in that space that makes them follow the rib cage. A hole in the wall or in the lung loses that pressure: the lung collapses (pneumothorax).

2. **Closed or open, the mechanism tells you what can break.** Blunt: rib fractures, pulmonary contusion, ruptured vessels, displaced heart, and often liver, spleen or kidneys beneath the lower ribs. Penetrating (ski pole, ice axe): air in the pleura, hemorrhage. The first complication to watch for is ineffective breathing.

3. **The abdomen: hollow organs and solid organs.** Solid organs (liver, spleen, kidneys) bleed: internal hemorrhage and shock. Hollow organs (stomach, intestines, bladder) perforate: contents in the cavity, infection, peritonitis. Both produce a painful, tense or rigid abdomen, and a patient who does not want to move.

4. **The four quadrants structure the physical exam.** A horizontal line and a vertical line crossing at the navel. Examine each quadrant separately: tenderness, guarding, rigidity, bruising, distension. Compare what the patient feels (subjective) with what you palpate (objective).

5. **In class: the injuries themselves.** Rib fractures and flail chest, open pneumothorax and occlusive dressing, tension pneumothorax, closed and open abdominal injuries, evisceration. This lesson gives you the map; the in-person session gives you the skills.

LEARNER'S MANUAL
Sections 8.8 and 8.9 (anatomy) · Online encyclopedia

FIELD MANUAL
Pages 124 to 127, chest and abdominal trauma

IN CLASS
Lung model, chest and abdominal injuries, rib fracture scenario